

Title page

Title

Empiria: A Modular-Synthesizer Platform for Simulation-Based Statistics Education

Short title (running head, ≤50 characters)

Empiria for Statistics Education

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Kevin Schoenholzer (sole author; details above).

Author contributions

K.S. is the sole author. He conceived the platform, designed and implemented the software (C++ for the plugins; Python for the patch generator), wrote the documentation and starter patches, and wrote the manuscript.

Funding

This work received no external funding.

Conflicts of interest

The author declares no competing interests. The platform is released under GPL-3.0 from the author's personal open-source imprint ("SHLabs"); the imprint is not a commercial entity and yields no income to the author.

Data and code availability

All source code, panels, documentation, three starter .vcv patches, and the Python patch generator are available under GPL-3.0 at:

<https://github.com/kevinschoenholzer/empiria>

The five plugins (Polis, Methods, Epi, Space, Decisions) are also bundled into VCV Rack .vcvplugin archives in the same repository for direct installation. A submission to the VCV Rack Library is in progress (see SUBMISSION.md in the repository for the cross-platform binary distribution checklist).

Ethical statement

The work does not involve human subjects, animal subjects, or secondary analysis of personal data. The synthetic-data examples in the paper are generated by deterministic seeded simulators within Empiria itself.

Acknowledgements

I thank the developers of VCV Rack for maintaining a free, open, and powerful environment on top of which Empiria is built, and the broader open-source modular-synthesizer plugin community whose public source code informed many of the implementation choices.

Word count

- Main text (excluding abstract, references, tables, figures): approximately 8,500 words.
- Abstract: 250 words.
- References: 36 entries with DOIs where available.
- Tables: 6 (module listing per plugin $\times$ 5 + workflow summary).
- Figures: TBD — to be added during typesetting from the platform's on-panel visualizations.

Manuscript category

Article (research paper).